

A PRELIMINARY PROJECT REPORT ON

HUMAN POSTURE ANALYSIS AND CORRECTION

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2024 -2025



CERTIFICATE

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ABSTRACT

This project presents an innovative approach to human posture analysis and correction, utilizing anthropometric data for enhanced precision. By integrating computer vision, and machine learning, the system captures and analyzes postural data in real-time. Accurate assessment of postural deviations and personalized corrective strategies are achieved through robust algorithms for feature extraction and pattern recognition. This approach not only improves posture and reduces discomfort but also significantly contributes to the prevention of musculoskeletal disorders, enhancing overall health and well-being. The system provides continuous monitoring and feedback, empowering users to make informed adjustments throughout the day. This comprehensive framework offers a promising solution for individuals and organizations aiming to promote better posture and prevent related health issues, with applications in workplaces, healthcare, and everyday life.

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1. INTRODUCTION

Even though maintaining an appropriate posture is essential for overall health and inclusion, sedentary activities and long-term sitting among others have been precursors to the bad posture claims. Wrong postures are known to lead to many painful musculoskeletal disorders that can affect an individual's day-to-day engagements. This has resulted in the need for new technologies, which are able to assess the person's positioning and suggest alterations in such a way that the user will not have health risks in the future.

In this project, an outstanding way of posture assessment, monitoring, and correction, is proposed based on the application of modern technologies, including, but not limited to, wearable sensors, computer vision, and machine learning. These technologies allow the collection and processing of posture data in a time effective manner and efficiently rendering postural evaluation.

The strategy is based on various measurements referred to as anthropometric that relate to a specific person in improving the effectiveness of the solution and providing effective corrective intervention. In addition to such features on posture defect detection, it also provides, in real time, very powerful pattern recognition and feature extraction algorithms for the user of the system.

Integrating constant tracking and feedback through this strategy leads to an active and knowledgeable process of relieving posture correction, improvement of comfort and lowering the occurrence of MSD.

1.1. MOTIVATION

As people are increasingly leading a sedentary lifestyle in this modern world, the long-term exposure of fixed postures has been noticed as one of the causes of several health disorders, including musculoskeletal disorders, chronic pain, and general deterioration of health. Moreover, as technology has advanced and more people are sedentary in front of devices for longer durations, the importance of good posture has increased in such management techniques.

The presence of today's posture correction therapy is not as dependable simply because it becomes limited to poor graphic design or too little use or total absence of associated feedback plans. Wearable sensors, computer vision, and machine learning are technological changes that can be applied in developing a unique system that also implements intervention along with detection of abnormalities in posture. This new approach has the objective of helping people achieve a higher quality of life over the long term by encouraging them to refrain from slumping and other poor postural habits.

1.2. PROBLEM DEFINITION

The problem addressed can be divided into two key components:

1. Real-Time Posture Detection and Analysis

The objective is to design a system that integrates computer vision techniques for real-time monitoring of human postures. The system will gather anthropometric information and use machine learning models to assess the differences from the ideal standing position. This part of the project seeks to assess the performance of efficient classification of different posture states such as correct, and forward head posture, or slouching using multi-source streamed data.

2. Corrective feedback and preventive alerts

This aspect is concerned with the provision of timely corrective feedback on the changes in posture that are beyond the allowed margin. The system will also notice such patterns and encourage users with appropriate suggestions. Furthermore, it shall also issue preventive cues such as the intent to perform or actual activity of remaining in a specific position for too long or repeating certain positions frequently. This will help in the prevention of work-related musculoskeletal disorders (MSD) by cutting down on the risks associated with these activities in the long run.

2. LITERATURE SURVEY

- I. Ko et al. (2024) proposed a real-time AI-based posture correction system for powerlifting exercises (bench press, squat, and deadlift) using YOLOv5 for person detection and MediaPipe for pose estimation. Their approach includes a MultiPose Exercise Dataset (MPED) containing 3D joint coordinates for concentric and eccentric phases. They trained machine learning and deep learning models to classify postures and provide detailed feedback. The system offers real-time corrective suggestions, improving exercise safety and performance, especially for high-risk, compound exercises often overlooked in previous research.
- II. Kim et al. in their paper analyzed a system for analysis of the posture aimed at identifying postural defects that utilizes computerized vision. The experts evaluate its compliance with self-evaluations, concentrating on the precision and dependability in detecting postural problems. The study demonstrates that incorporation of automated evaluation of posture may improve clinical management by providing measures in time to avert problems associated with poor posture.
- III. The article by Luangaphirom et al. presents the web application, created for monitoring the correct posture in weightlifting. It is concentrated on fundamental bodyweight movements. The application uses a live camera along with MediaPipe and OpenCV for analyzing the users' body position. The application uses the calculated angles and provides real-time feedback on whether the user meets the criteria in all parameters for the proper repetition. The testing results show that the developed application has great potential for further testing.

- IV. Sakshi et al. examines the recent developments in computer vision technology that focus on the automatic recognition of human actions in videos and images, with the goal of enhancing posture through detailed reports. The proposed model provides corrective feedback for yoga poses both in pre-recorded videos and in real-time, achieving an accuracy of 93.43%. Additionally, the paper offers a review of prior pose estimation methods and elaborates on the various deep learning models employed for the classification of yoga poses.
- V. The author, Debnath B et al., emphasizes that the field of computer vision has made significant advancement in the analysis of human motion, particularly in areas such as pose estimation, gesture recognition, and activity recognition. Applications aimed at monitoring functional rehabilitation for individuals with musculoskeletal or physical disabilities necessitate a comparative assessment of human motion. This survey compiles essential literature on vision-based monitoring and physical rehabilitation providing an overview of the current research landscape in this domain.
- VI. In their paper, Li B. et al., introduces a new hybrid method for identifying human posture. This technique combines anthropometric measures, depth data, skeletal data, and a backpropagation neural network (BPNN) with Kinect V2. The method divides postures into four distinct categories: standing, sitting or kneeling, sitting cross-legged, and others. The studies' outcomes show how reliable and successful this hybrid strategy is, with an average recognition accuracy of 99.09%.
- VII. Raju J.B.V. et al., in paper aim to present a smart posture recognition and correction system. Sitting in the wrong posture for a long period of time results in many health problems such as poor circulation, back pain, cervical pains, soreness and also decrease in eyesight in the long run. The model proposed makes use of real time skeletal points extraction. This system is based on CV and ML algorithms.

- VIII. Kuppusamy. P et al. examines the latest developments in artificial intelligence that improve environmental learning, object recognition and sequence prediction. The advent of embedded technology has led to a decrease in the costs associated with surveillance systems, which are capable of capturing and storing environmental data. This study investigates the use of video for the identification of human actions and behaviors, suggesting integration of convolutional neural networks (CNN) and long short-term memory (LSTM) recurrent neural networks for video analysis.
- IX. A. Toshev et al. proposed a seminar paper that introduces DeepPose, a deep learning framework for accurate human pose estimation, foundational for posture analysis and correction systems.
- X. F. Rishan et al, highlights the growing trend of yoga, which is linked to its various physical, mental, and spiritual benefits. The developed system aims to identify user poses and provide immediate visual assistance, thus improving user engagement. The pose detection component has been trained to identify six unique asanas, attaining an impressive accuracy rate of 99.91% using OpenPose-based pose estimation.
- XI. W. Ni et al. presents a study on multi-node sensors aimed at detecting human posture, emphasizing innovative approaches in computer vision and communication technologies. It collects kinematic parameters of the human body to investigate the principles that underlie human movement. This research explores the use of multi-carrier modulation in humans.
- XII. Petermann X. B. et al. discusses various approaches to measure body posture in physiotherapy research. Through a reviewed literature, ten different techniques of postural assessment have been identified, such as Postural Assessment Software (PAS), digital photography, photogrammetry, standard photography, and others. From the result, it can be deduced that PAS is a leading technique in the several methods used in body postural assessment because it was the most opted for technique by many scientists.

- XIII. Rajendran A.K et al. in study investigates the influence that yoga represents as a form of physical activity and its applications in personal care health. Adequate body postural posture must be in perfect alignment while performing yogic asanas. Advances in algorithms and sensors in computer vision have given a possibility to predict yoga postures and analyze movements automatically. This study conducts a general analysis of yoga posture identification systems including machine learning, deep learning and computer vision.
- XIV. Jiang X. et al. in paper introduces recent methodologies concerning posture recognition and review recently developed techniques and algorithms, like scale-invariant feature transformation, support vector machines (SVM), Gaussian mixture models, dynamic time warping, hidden Markov models (HMM), and convolutional neural networks (CNN).The paper further discusses advanced neural networks in posture recognition, including graph neural networks, ensemble learning, transfer learning and explainable deep neural networks. The study emphasizes a critical need for 3D benchmark datasets.
- XV. Ding W. et al. discusses the fundamental question of research using skeleton data on human posture recognition in human-computer interaction. In order to better enhance human posture recognition, the authors here introduce a new algorithm that makes use of multiple features and rule learning.The authors use rule learning combined with bagging and methods from the random subspace to generate different kinds of samples and features. This algorithm was tested on 4 datasets to classify human postures, and experimental results show it can make a distinction between them.

- XVI. Piñero-Fuentes E. et al. describes how the pandemic shifted face-to-face workplaces to teleworking, which in many ways has led to spending too much time in front of computers. It provides recommendations for preventing health issues attributed to poor posture. Results are given in the form of video processing by the system up to 25 frames per second with less power consumption below 10 watts yet reaching above 80% accuracy in the detection of postural patterns. This innovational approach represents a significant leap in occupational health tools designed to enhance worker ergonomics in the teleworking arena.

3. PROJECT PLAN

3.1 ESTIMATES

3.1.1. Initial Estimates

The project will be completed in phases, and the following time and resource estimates are made for each phase of development:

- **Phase 1: Literature Review & Requirement Gathering** (2 weeks)

Review current research on posture correction using video-based analysis. Gather requirements for video capture techniques, machine learning model selection, and system architecture.

- **Deliverable:** Literature review document and project requirements specification.

- **Phase 2: System Design** (3 weeks)

Design the high-level system architecture. Define the video capture module, frame preprocessing, machine learning-based posture detection, and feedback generation. Decide on model architecture, training methodology, and hardware requirements (e.g., camera and processing specifications).

- **Deliverable:** System architecture diagram, technical design document.

- **Phase 3: Data Collection & Preprocessing** (5 weeks)

The system will use publicly available datasets of human postures or capture custom videos for training and testing. Preprocessing will include frame extraction, resizing, and normalization. Additional steps may include augmenting the data for robustness.

- **Deliverable:** Preprocessed dataset ready for model training.

- **Phase 4: Model Development & Training** (6 weeks)

Build and train machine learning models (e.g., CNNs for frame-based analysis). Models will be trained to detect different postures from video frames and identify improper posture.

- **Deliverable:** Trained posture detection model, model evaluation report.

- **Phase 5: System Integration** (4 weeks)

Integrate the trained models into the application that processes video or real-time frames. Build a user interface for capturing videos, send frames to the model, and display feedback.

- **Deliverable:** Integrated posture analysis system with a functional UI.

- **Phase 6: Testing & Validation** (4 weeks)

Perform unit and system testing to ensure correct functionality. User testing will be conducted to gather feedback on usability and system performance.

- **Deliverable:** Test report and user feedback.

3.1.2. Reconciled Estimates

After refining the project scope, the following adjustments have been made:

- **Literature Review & Requirement Gathering:** Reduced to 1 week, as familiarity with video-based analysis allows for faster research.
- **System Design:** Kept at 3 weeks due to the complexity of designing a video-processing system.
- **Data Collection & Preprocessing:** Reduced to 4 weeks by leveraging publicly available posture datasets, speeding up the data collection process.
- **Model Development & Training:** Extended to 7 weeks to accommodate the need for optimizing real-time posture detection models.
- **System Integration:** Remains at 4 weeks, as integration and UI development follow typical timelines.
- **Testing & Validation:** Maintained at 4 weeks to ensure comprehensive testing and feedback incorporation.

3.2 RISK MANAGEMENT

3.2.1. Risk Identification

The primary risks identified for this project are:

- **Video Quality and Frame Rate Issues:** Low-quality video or frame rate drops may lead to inaccurate posture detection. Real-time frame capture could also be affected by hardware limitations.
- **Model Accuracy and Real-Time Performance:** Ensuring that the posture detection model is both accurate and efficient in real-time processing could be challenging.
- **User Interface Complexity:** Developing a simple yet effective interface for video capture, frame submission, and feedback delivery could be difficult, especially in real-time applications.

3.2.2 Risk Analysis

- **Video Quality:** The system's performance is highly dependent on the quality of the video. If the frame rate drops or the video is blurry, the model may not be able to detect postures accurately.
- **Model Accuracy:** Accuracy is a high-risk factor. The model must not only be able to identify static postures but also correct dynamic postures in real-time, which can be computationally intensive.
- **User Interface:** The interface must be intuitive, or users may not engage with the system effectively.

3.2.3 Overview of Risk Mitigation, Monitoring, Management

- **Video Quality:** We will recommend minimum camera resolution and frame rate requirements for optimal performance. Additionally, preprocessing steps such as filtering will be used to enhance video quality.
- **Model Optimization:** To handle real-time processing, we will implement optimizations like model pruning and quantization to reduce the computational load.
- **User Interface Feedback:** We will conduct early-stage user testing to gather feedback and make continuous improvements to the interface.

3.3 PROJECT SCHEDULE

3.3.1 Project Task Set

The project is broken into six major tasks:

1. **Literature Review & Requirement Gathering**
2. **System Design**
3. **Data Collection & Preprocessing**
4. **Model Development & Training**
5. **System Integration**
6. **Testing & Validation**

3.3.2 Task Network

The tasks follow a sequential order with some overlap between data collection and model development:

1. **Literature Review** → 2. **System Design** → 3. **Data Collection** → 4. **Model Development** → 5. **System Integration** → 6. **Testing & Validation**.

3.3.3 Timeline Chart

The timeline for the project is as follows:

- **Week 1-2:** Literature review and requirement gathering.
- **Week 3-5:** System design, including architecture and module definitions.
- **Week 6-9:** Data collection and preprocessing.
- **Week 10-16:** Machine learning model development and training.
- **Week 17-20:** System integration and user interface development.
- **Week 21-24:** Testing and validation.

3.4 TEAM ORGANIZATION

3.4.1 Team Structure

The project team is organized based on expertise and task specialization:

- **Krishna Patil:** Project leader, responsible for overall system design, project management, and ensuring timely delivery.
- **Prajwal Billade:** Focuses on machine learning model development and evaluation.
- **Samraddhi Rathor:** Handles data preprocessing, video capture module, and model integration.
- **Maitreyee Joshi:** Responsible for building the user interface and ensuring that the real-time feedback system is functional.

3.4.2 Management Reporting and Communication

- **Internal Communication:** Weekly meetings will be held to discuss progress. Each member will report their progress, and any roadblocks will be discussed and resolved.
- **External Communication:** Regular updates will be shared with project supervisors, including progress reports and feedback on technical design and implementation phases.

4. SOFTWARE REQUIREMENTS SPECIFICATION

4.1. INTRODUCTION

The **Human Posture Analysis and Correction** system uses computer vision and machine learning models to detect and correct human posture in real-time. The system captures video or frame data from a camera, processes the frames to extract human posture information, and provides corrective feedback to the user.

4.1.1. Project Scope

The system is intended to assist users in improving posture, reducing the risk of musculoskeletal issues. This system will be applicable in various fields, including:

- **Healthcare:** For rehabilitation and posture correction.
- **Workplace Ergonomics:** Monitoring posture for desk workers.
- **Fitness:** Ensuring proper form during exercises.

4.1.2. Use Cases and Characteristics

- **Use Case 1: Real-time Posture Correction**

A user sets up the camera, and the system captures live video. The system processes the frames and detects the user's posture. If improper posture is detected, the system provides corrective feedback in real-time.

- **Use Case 2: Posture Analysis from Pre-recorded Video**

Users can upload a pre-recorded video, which the system processes by extracting frames. It analyzes the frames for postural issues and provides feedback.

4.1.3. Assumptions and Dependencies

1. **Assumptions:**

- **Camera Quality:** It is assumed that the users will have access to a camera with sufficient resolution (at least 720p) and frame rate (30 FPS or higher) for accurate posture detection.
- **Adequate Lighting:** The environment in which the system will be used has adequate lighting to ensure clear video capture for better posture recognition.
- **User Device Capabilities:** It is assumed that users will be using devices (e.g., laptops or PCs) that meet the minimum hardware requirements for running the posture detection system efficiently.

2. Dependencies:

- **Machine Learning Model Accuracy:** The accuracy of the posture detection depends on the quality and diversity of the dataset used for training the model. The model should be pre-trained or fine-tuned on a comprehensive dataset.
- **Video Capture Library:** The system is dependent on libraries like **OpenCV** for capturing and processing video frames. Any bugs or performance issues in these libraries could impact the overall system performance.
- **Real-time Processing Capability:** The system depends on the processing power of the user's device. High frame rates and real-time analysis require sufficient CPU/GPU power.

4.2. FUNCTIONAL REQUIREMENTS

4.2.1. System Feature 1: Video Capture and Frame Processing

- The system will allow users to connect a camera or upload a video file.
- Frames will be extracted from the video at a specified frame rate for analysis.
- Each frame will undergo preprocessing (e.g., resizing, normalization) to prepare it for the model.

4.2.2. System Feature 2: Posture Detection using Machine Learning

- The system will use a trained machine learning model (e.g., CNN) to analyze frames and detect the user's posture.
- The model will classify postures into correct or incorrect categories based on predefined metrics.

4.2.3. System Feature 3: Real-time feedback mechanism

- If incorrect posture is detected, the system will provide corrective feedback in real-time through visual or audio cues.
- The feedback will suggest corrective actions or adjustments to the user's posture.

4.3. EXTERNAL INTERFACE REQUIREMENTS

4.3.1. User Interfaces

- **Video Input Interface:** The system will provide a user-friendly interface for capturing live video from a camera or uploading a video file for analysis.
- **Feedback Interface:** Visual feedback will be provided through on-screen prompts.

4.3.2. Hardware Interfaces

- The system will require access to a camera for live video capture. The minimum required camera specifications include:
 - **Resolution:** 720p (1280x720) minimum for effective posture detection.
 - **Frame Rate:** 30 FPS minimum to ensure smooth posture detection.

4.3.3. Software Interfaces

- **Operating System:** The application will be compatible with Windows, macOS, and Linux operating systems.

- **Development Framework:** The system will utilize Python, OpenCV for video processing, and TensorFlow for machine learning.

4.4. NON FUNCTIONAL REQUIREMENTS

4.4.1. Performance Requirements

- The system must process frames in real-time with a latency of no more than 100 milliseconds between frame capture and feedback delivery.
- The posture detection model should have an accuracy of at least 90% in detecting incorrect postures.

4.4.2. Safety Requirements

- The system should ensure that no user data, especially video recordings, are shared without explicit consent.
- Feedback should be clear and actionable to avoid causing user discomfort during correction.

4.4.3. Security Requirements

- User video data must be encrypted during storage and transmission to protect privacy.
- Access control mechanisms should be in place to ensure that only authorized users can view or modify their posture data.

4.5. SYSTEM REQUIREMENTS

4.5.1. Database Requirements

- The system will use a lightweight database (e.g., SQLite or MongoDB) to store user data, including past analyses, reports, and feedback history.

4.5.2. Software Requirements(Platform choice)

- **Programming Languages:** Python 3.x.
- **Libraries/Frameworks:**
 - **OpenCV** for video capture and frame processing.

- **TensorFlow/Keras** for building and running machine learning models.
- **Flask/Django** for creating the web-based or desktop application interface.

4.5.3. Hardware Requirements

- **Minimum Requirements:**
 - Processor: Intel Core i5 or equivalent.
 - RAM: 8 GB.
 - Storage: 500 GB HDD or SSD for storing videos and reports.
 - Camera: 720p resolution with at least 30 FPS for video capture.

4.6. ANALYSIS MODELS : SDLC MODEL TO BE APPLIED

The project will follow an **Agile Software Development Life Cycle (SDLC)** model, allowing for iterative development and continuous feedback from users. Key stages include:

- **Planning & Requirement Gathering:** Initial research, stakeholder interviews, and defining project scope.
- **Design:** High-level design of the video capture and posture analysis modules.
- **Development:** Coding and implementing machine learning models, followed by UI development.
- **Testing:** Both unit and integration testing of the system with real-time video capture.
- **Deployment:** Rolling out the system to end-users for feedback and adjustments.

5. SYSTEM DESIGN

5.1. SYSTEM ARCHITECTURE

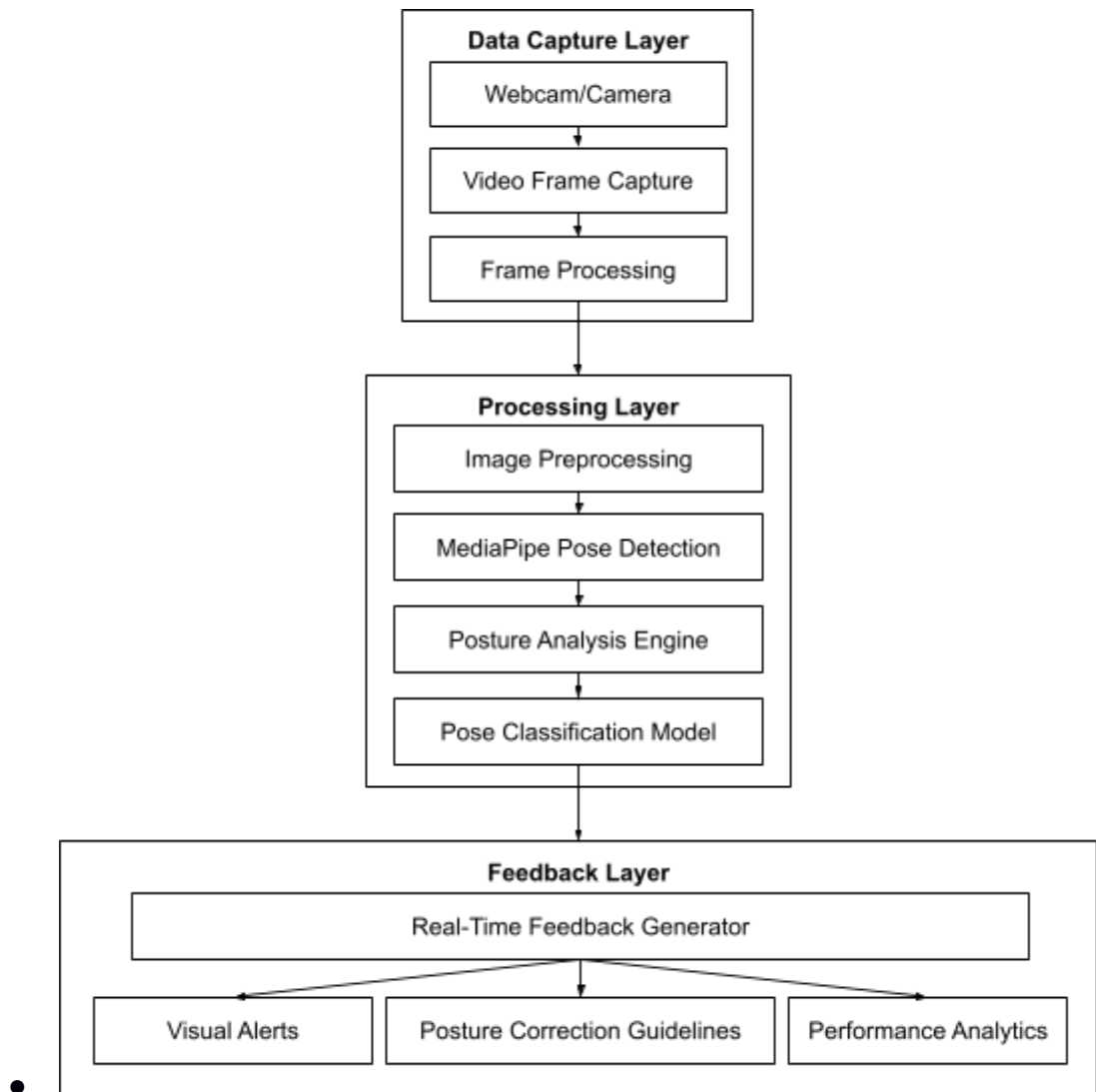


Figure 5.1 Architecture Diagram

5.2. ENTITY RELATIONSHIP DIAGRAM

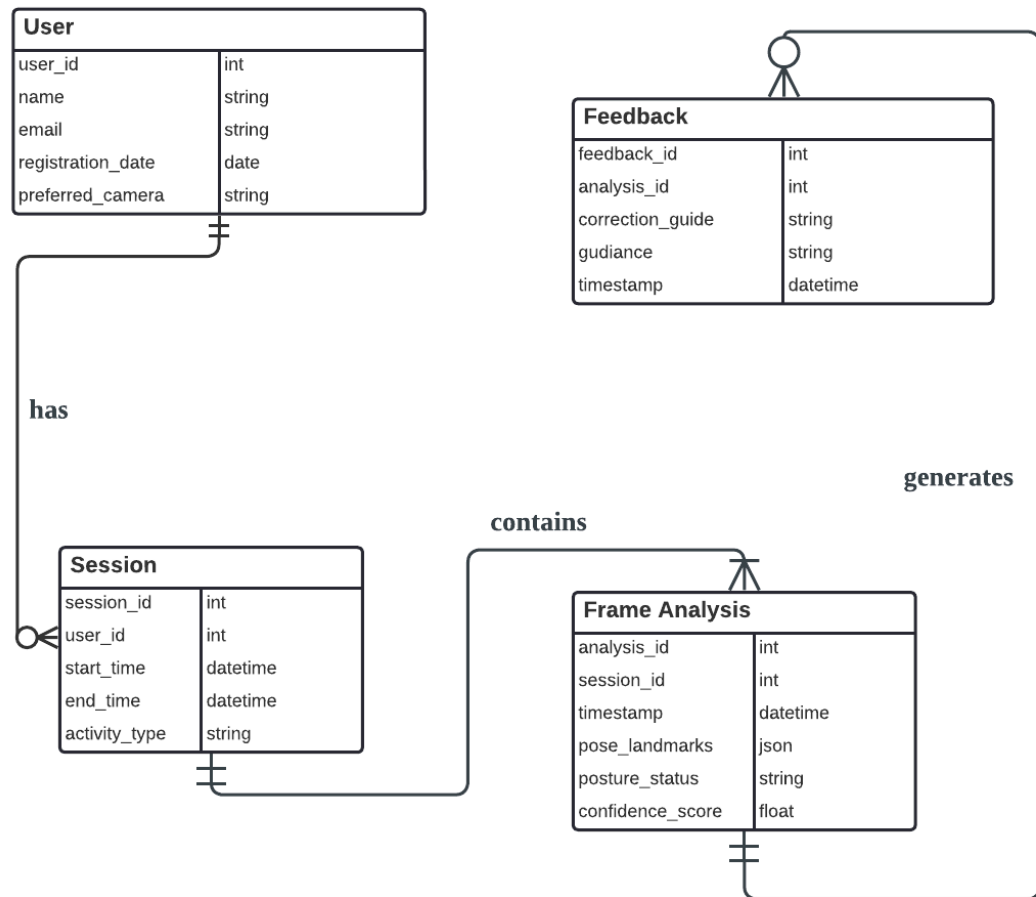


Figure 5.2 ER Diagram

5.3. UML DIAGRAMS

5.3.1. Use Case Diagram

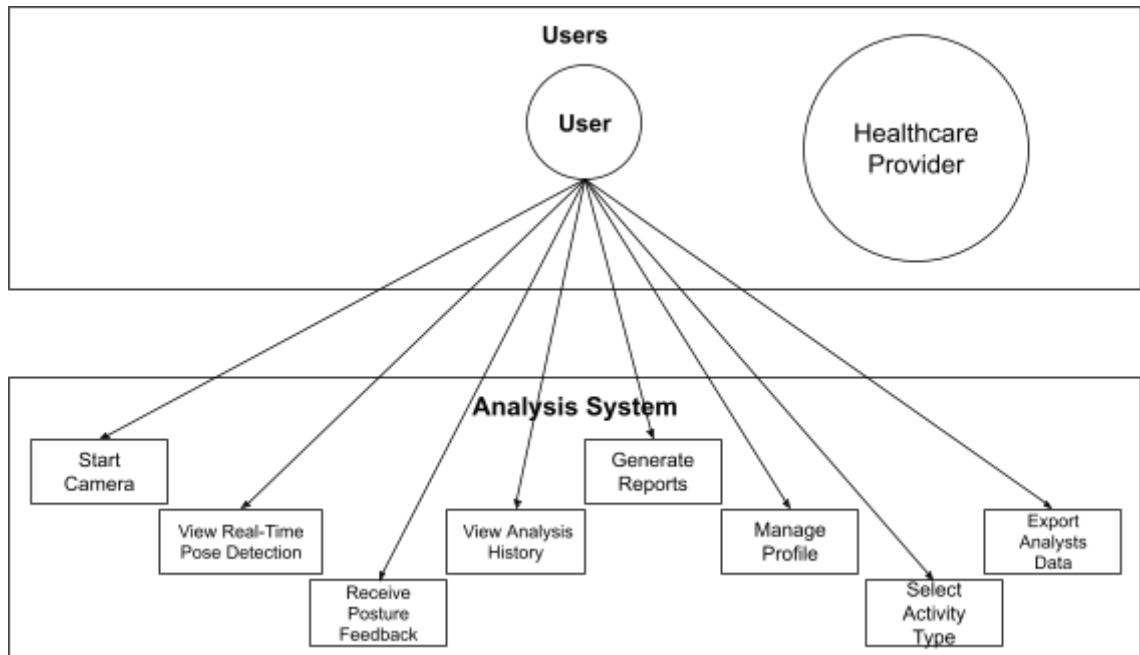


Figure 5.3 Use Case Diagram

5.3.2. Flow Chart Diagram

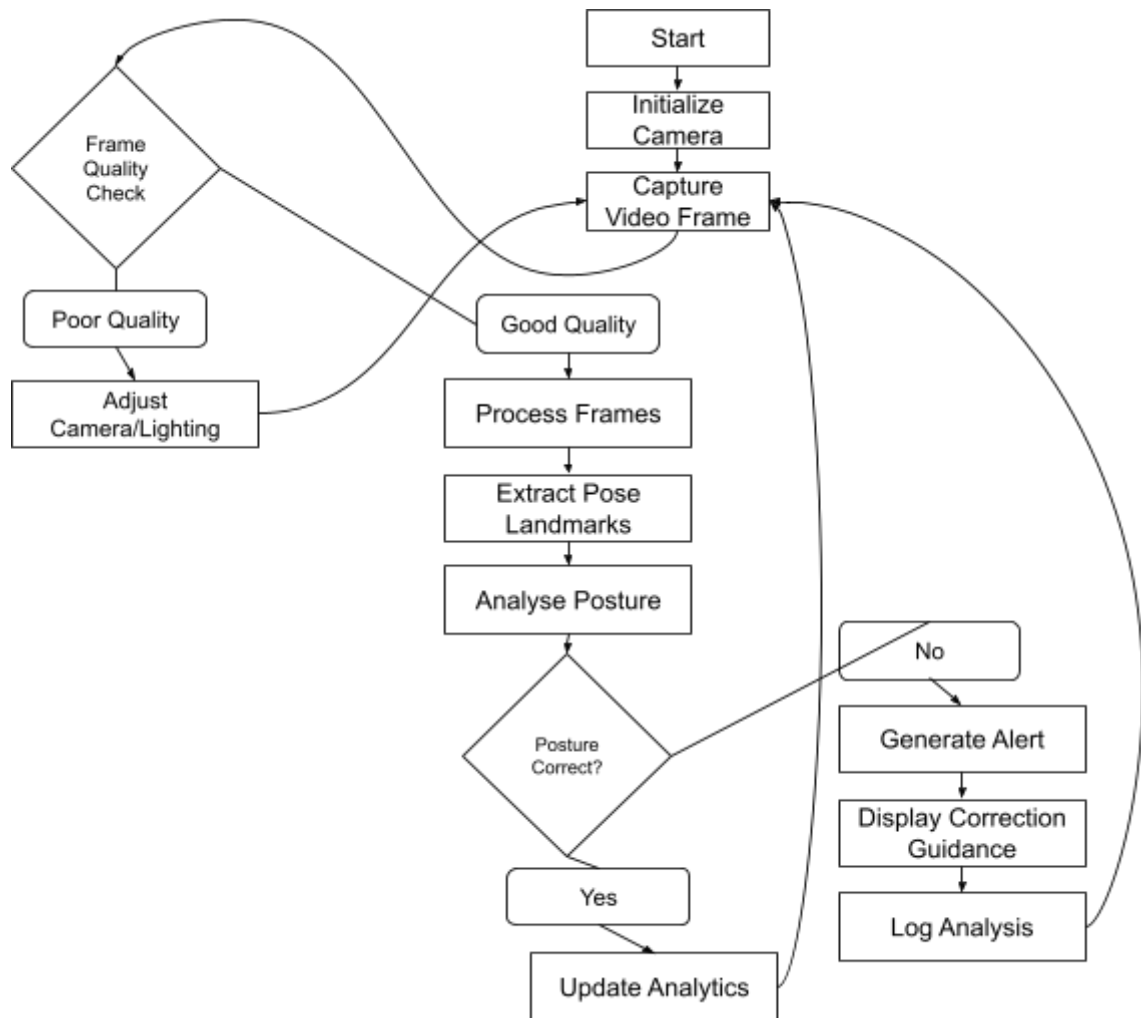


Figure 5.4 Flow Chart Diagram

5.3.3 Data Flow Diagram

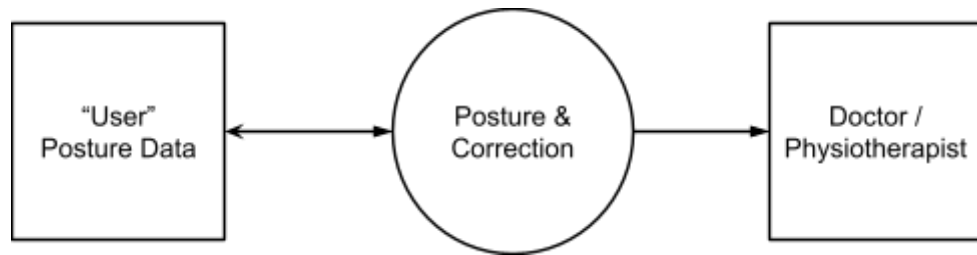


Figure 5.5 Data Flow Diagram

6. OTHER SPECIFICATIONS

6.1. ADVANTAGES

1. **Real Time Feedback:** Users are promptly alerted about their posture, allowing for immediate adjustments and minimizes the likelihood of developing musculoskeletal disorders.
2. **Non-Invasive Evaluation:** This system allows for the evaluation of posture without the necessity for physical examinations or interventions, thus ensuring a more comfortable experience for users.
3. **Continuous Monitoring:** The incorporation of machine learning enables ongoing surveillance, thereby enhancing the accuracy of posture evaluations over time.
4. **Versatility:** The system is adaptable for use in diverse environments, such as workplaces and healthcare settings.
5. **Economical Approach:** By automating the posture assessment process, the system reduces the frequency of required visits to healthcare providers, resulting in savings in both time and resources.

6.2. LIMITATIONS

1. **Reduced Accuracy in Complex Conditions:** The effectiveness of posture detection can be compromised in scenarios where the system struggles, such as in dimly lit spaces or when the line of sight is obstructed.
2. **Dependency on High-Quality Equipment:** The system requires high-end cameras and sensors, which may not be affordable by certain users.
3. **Lack of Customization:** In the absence of appropriate adjustments, the system may fail to sufficiently address individual anatomical differences or specific health concerns.
4. **Privacy Issues:** Ongoing data collection and surveillance may lead to privacy concerns, especially in sensitive settings such as residential or workplace environments.

6.3. APPLICATIONS

- 1. Healthcare and Rehabilitation:** It provides a check and the corrective change of posture in patients recovering from musculoskeletal disorders or injuries.
- 2. Ergonomics in the Workplace:** This approach can assist organizations in correcting the postures of employees, reducing their fatigue levels, and eliminating chronic health disorders characterized by adverse habits of sitting or standing.
- 3. Sports and Fitness:** This method can be applied to sports and fitness athletes and trainers. It can help maintain optimal postures during training, thus improving the performance levels and lowering the possibilities of injuries that may result from other less healthy postures.
- 4. Personal Use:** For individual use, it could be used by persons looking to obtain healthy bases for their daily posture, thereby achieving better health in general.

7. RESULTS

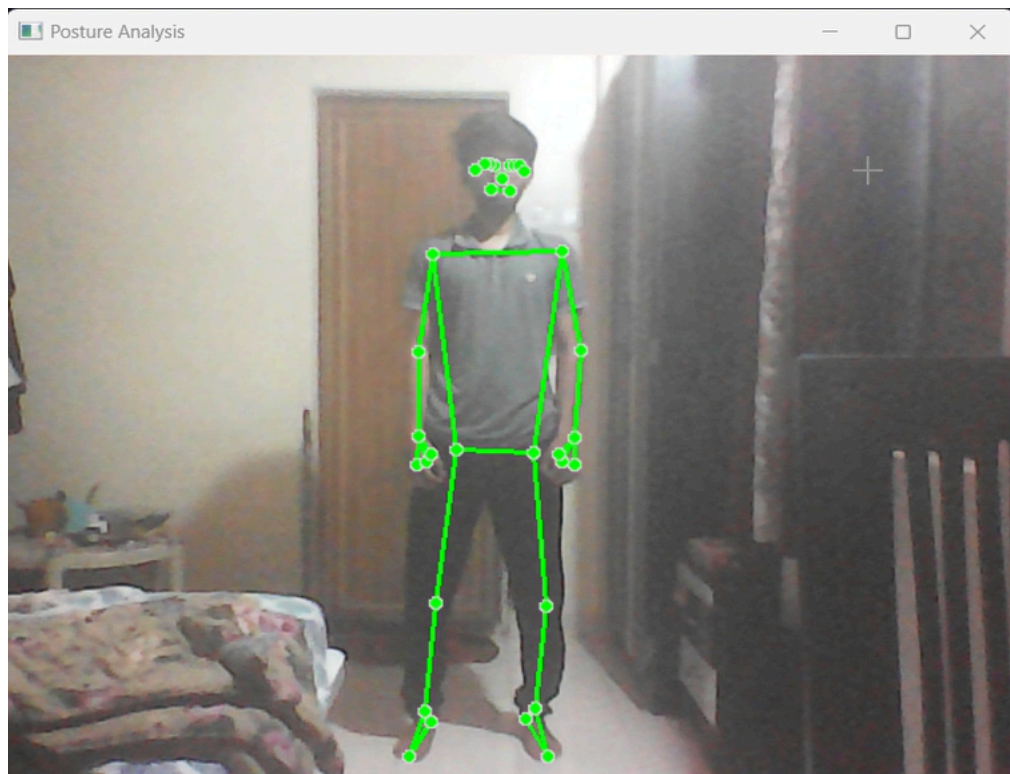


Figure 7.1 Standing posture detected using MediaPipe

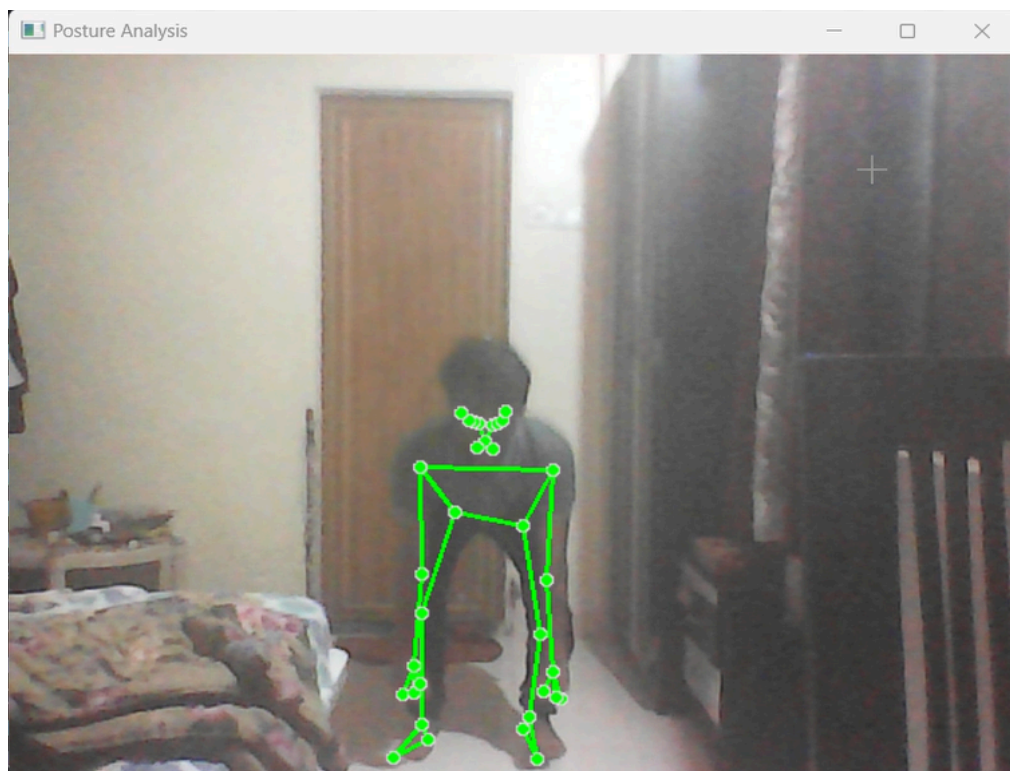


Figure 7.2 Squatting posture detected using MediaPipe

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

INFO:      127.0.0.1:54560 - "POST /process-frame HTTP/1.1" 200 OK
INFO:      127.0.0.1:54565 - "POST /process-frame HTTP/1.1" 200 OK
INFO:      127.0.0.1:54568 - "POST /process-frame HTTP/1.1" 200 OK
INFO:      127.0.0.1:54570 - "POST /process-frame HTTP/1.1" 200 OK
INFO:      127.0.0.1:54572 - "POST /process-frame HTTP/1.1" 200 OK
INFO:      127.0.0.1:54574 - "POST /process-frame HTTP/1.1" 200 OK
INFO:      127.0.0.1:54576 - "POST /process-frame HTTP/1.1" 200 OK
INFO:      127.0.0.1:54580 - "POST /process-frame HTTP/1.1" 200 OK
2025-04-16 19:20:40.642595: E tensorflow/core/util/util.cc:131] oneDNN supports DT_HALF only on platforms with AVX-512. Falling back to the default Eigen-based implementation if present.
1/1 ██████████ 19s 19s/step
DEBUG:root:MODEL PREDICTION: correct
INFO:      127.0.0.1:54582 - "POST /process-frame HTTP/1.1" 200 OK
1/1 ██████████ 1s 1s/step
DEBUG:root:MODEL PREDICTION: leaning_forward
INFO:      127.0.0.1:54584 - "POST /process-frame HTTP/1.1" 200 OK
1/1 ██████████ 1s 1s/step
DEBUG:root:MODEL PREDICTION: leaning_forward
INFO:      127.0.0.1:54586 - "POST /process-frame HTTP/1.1" 200 OK
1/1 ██████████ 1s 1s/step
DEBUG:root:MODEL PREDICTION: correct
INFO:      127.0.0.1:54589 - "POST /process-frame HTTP/1.1" 200 OK
1/1 ██████████ 1s 1s/step
DEBUG:root:MODEL PREDICTION: correct
INFO:      127.0.0.1:54593 - "POST /process-frame HTTP/1.1" 200 OK
1/1 ██████████ 4s 4s/step
DEBUG:root:MODEL PREDICTION: bar_too_far
INFO:      127.0.0.1:54713 - "POST /process-frame HTTP/1.1" 200 OK
1/1 ██████████ 2s 2s/step
DEBUG:root:MODEL PREDICTION: bar_too_far
INFO:      127.0.0.1:54715 - "POST /process-frame HTTP/1.1" 200 OK
1/1 ██████████ 2s 2s/step
```

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

1/1 ██████████ 2s 2s/step
DEBUG:root:MODEL PREDICTION: rounding_lower_back
INFO:      127.0.0.1:54717 - "POST /process-frame HTTP/1.1" 200 OK
1/1 ██████████ 2s 2s/step
DEBUG:root:MODEL PREDICTION: rounding_lower_back
INFO:      127.0.0.1:54719 - "POST /process-frame HTTP/1.1" 200 OK
1/1 ██████████ 2s 2s/step
DEBUG:root:MODEL PREDICTION: bar_too_far
INFO:      127.0.0.1:54723 - "POST /process-frame HTTP/1.1" 200 OK
1/1 ██████████ 2s 2s/step
DEBUG:root:MODEL PREDICTION: correct
INFO:      127.0.0.1:54725 - "POST /process-frame HTTP/1.1" 200 OK
1/1 ██████████ 2s 2s/step
DEBUG:root:MODEL PREDICTION: correct
INFO:      127.0.0.1:54727 - "POST /process-frame HTTP/1.1" 200 OK
1/1 ██████████ 2s 2s/step
DEBUG:root:MODEL PREDICTION: correct
INFO:      127.0.0.1:54729 - "POST /process-frame HTTP/1.1" 200 OK
1/1 ██████████ 2s 2s/step
DEBUG:root:MODEL PREDICTION: leaning_forward
INFO:      127.0.0.1:54733 - "POST /process-frame HTTP/1.1" 200 OK
1/1 ██████████ 2s 2s/step
DEBUG:root:MODEL PREDICTION: rounding_lower_back
INFO:      127.0.0.1:54736 - "POST /process-frame HTTP/1.1" 200 OK
1/1 ██████████ 2s 2s/step
DEBUG:root:MODEL PREDICTION: rounding_lower_back
```

```
INFO:      127.0.0.1:54740 - "POST /process-frame HTTP/1.1" 200 OK
1/1 ██████████ 2s 2s/step
DEBUG:root:MODEL PREDICTION: leaning_forward
INFO:      127.0.0.1:54742 - "POST /process-frame HTTP/1.1" 200 OK
1/1 ██████████ 2s 2s/step
DEBUG:root:MODEL PREDICTION: rounding_lower_back
INFO:      127.0.0.1:54744 - "POST /process-frame HTTP/1.1" 200 OK
1/1 ██████████ 2s 2s/step
DEBUG:root:MODEL PREDICTION: correct
INFO:      127.0.0.1:54746 - "POST /process-frame HTTP/1.1" 200 OK
1/1 ██████████ 2s 2s/step
DEBUG:root:MODEL PREDICTION: bar_too_far
INFO:      127.0.0.1:54751 - "POST /process-frame HTTP/1.1" 200 OK
1/1 ██████████ 2s 2s/step
DEBUG:root:MODEL PREDICTION: correct
INFO:      127.0.0.1:54753 - "POST /process-frame HTTP/1.1" 200 OK
1/1 ██████████ 2s 2s/step
DEBUG:root:MODEL PREDICTION: bar_too_far
```

8. CONCLUSION AND FUTURE WORK

8.1. CONCLUSION

The computer vision-based posture analysis system, despite incorporating numerous innovative technologies, serves as a crucial first step towards posture correction and the prevention of musculoskeletal disorders. It utilizes real-time data from wearable sensors, making machine learning relevant for this purpose. The advantages of this system are almost exclusively automation, access, and personalized feedback. Its potential is very attractive for various kinds of applications, ranging from healthcare to workplace ergonomics and personal well-being. However, the challenge of dependency on advanced hardware is mainly concerning the environment and the private data that it shares among its users. This system has the potential to drastically change the approach of health care in regards to long-term conditions pertaining to posture while minimizing discomfort, maximizing productivity, and increasing well-being.

Further developments towards high precision and flexibility of this system may eventually find their way as an extremely general solution contributing to better health, reduction in musculoskeletal disorders, and improvement in quality of life across different environments and user groups.

8.2. FUTURE WORK

- **Increased Accuracy under Challenging Conditions:** Ensure better posture detection with low light conditions or obstructions in direct line of sight.
- **Personalization:** Develop models customized for particular users and their anatomy and specific health conditions for accurate feedback.
- **Privacy and Security:** Design data with stronger security measures into place to protect privacy while ensuring continuous monitoring.
- **Integration with Mobile and Wearable Technologies:** Expand the compatibility of mobile and wearable technologies to support a system for real-time posture correction.
- **Clinical Applications:** The system can be extended to support rehabilitation therapies whose aim is the recovery from injuries as well as the treatment of musculoskeletal conditions.

This future work aims to improve the system's capabilities, making it more versatile and effective in real-world applications.

9. REFERENCES

1. Ko, Y., Nasridinov, A., & Park, S. (2024). Real-Time AI posture correction for powerlifting exercises using YOLOV5 and MediaPipe. *IEEE Access*, 12, 195830–195853. <https://doi.org/10.1109/access.2024.3516723>
2. Kim, K. H., Sohn, M., & Park, C. G. (2022). Conformity assessment of a computer vision-based posture analysis system for the screening of postural deformation. *BMC Musculoskeletal Disorders*, 23(1). <https://doi.org/10.1186/s12891-022-05742-7>
3. Luangaphirom, T., Lueprasert, S., Kaewvichit, P., Boonphotsiri, S., Burapasikarin, T., & Siriborvornratanakul, T. (2024). Real-time weight training counting and correction using MediaPipe. *Advances in Computational Intelligence*, 4(2). <https://doi.org/10.1007/s43674-024-00070-w>
4. Sakshi, N., & Saini, S. (2024). Yoga Posture Estimation and Correction using Mediapipe and Deep Learning Models. In *Lecture notes in networks and systems* (pp. 517–529). https://doi.org/10.1007/978-981-99-7862-5_39
5. Debnath, B., O'Brien, M., Yamaguchi, M., & Behera, A. (2021). A review of computer vision-based approaches for physical rehabilitation and assessment. *Multimedia Systems*, 28(1), 209–239. <https://doi.org/10.1007/s00530-021-00815-4>
6. Li, B., Han, C., & Bai, B. (2019). Hybrid approach for human posture recognition using anthropometry and BP neural network based on Kinect V2. *EURASIP Journal on Image and Video Processing*, 2019(1). <https://doi.org/10.1186/s13640-018-0393-4>
7. Raju, J. B. V. P., Reddy, Y. C., & G, P. R. (2019). Smart Posture Detection and Correction System Using Skeletal Points Extraction. In *Learning and analytics in intelligent systems* (pp. 177–181). https://doi.org/10.1007/978-3-030-24322-7_23
8. Pothanaicker, K. & VIT-AP University. (2019). Human Action Recognition using CNN and LSTM-RNN with Attention Model. *International Journal of Innovative Technology and Exploring Engineering (IJITEE)*, 1639–1640. <https://www.researchgate.net/publication/334508614>
9. Toshev, A., & Szegedy, C. (2014). DeepPose: Human Pose Estimation via Deep Neural Networks. *2014 IEEE Conference on Computer Vision and Pattern Recognition*. <https://doi.org/10.1109/cvpr.2014.214>

10. Rishan, F., De Silva, B., Alawathugoda, S., Nijabdeen, S., Rupasinghe, L., & Liyanapathirana, C. (2020). Infinity Yoga Tutor: Yoga Posture Detection and Correction System. *2020 5th International Conference on Information Technology Research (ICITR)*.
<https://doi.org/10.1109/icitr51448.2020.9310832>
11. Ni, W., Gao, Y., Lucev, Z., Pun, S. H., Cifrek, M., Vai, M. I., & Du, M. (2016). Human posture detection based on human body communication with muti-carriers modulation. *2016 39th International Convention on Information and Communication Technology, Electronics and Microelectronics (MIPRO)*.
<https://doi.org/10.1109/mipro.2016.7522151>
12. Petermann, X. B., & Meereis, E. C. W. (2016). Postura corporal: Uma revisão sistemática sobre métodos de avaliação. *Manual Therapy Posturology & Rehabilitation Journal*, 14, 273.
<https://doi.org/10.17784/mtprehabjournal.2016.14.273>
13. Rajendran, A. K., & Sethuraman, S. C. (2023). A Survey on Yogic Posture Recognition. *IEEE Access*, 11, 11183–11223.
<https://doi.org/10.1109/access.2023.3240769>
14. Jiang, X., Hu, Z., Wang, S., & Zhang, Y. (2023). A survey on Artificial Intelligence in Posture Recognition. *Computer Modeling in Engineering & Sciences*, 137(1), 35–82.
<https://doi.org/10.32604/cmes.2023.027676>
15. Ding, W., Hu, B., Liu, H., Wang, X., & Huang, X. (2020). Human posture recognition based on multiple features and rule learning. *International Journal of Machine Learning and Cybernetics*, 11(11), 2529–2540.
<https://doi.org/10.1007/s13042-020-01138-y>
16. Piñero-Fuentes, E., Canas-Moreno, S., Rios-Navarro, A., Domínguez-Morales, M., Sevillano, J. L., & Linares-Barranco, A. (2021). A Deep-Learning Based Posture Detection System for Preventing Telework-Related Musculoskeletal Disorders. *Sensors*, 21(15), 5236.
<https://doi.org/10.3390/s21155236>